

PEI-YU LIN

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EDUCATION

RWTH Aachen University, Aachen, Germany

M.Sc. in Electrical Engineering

Apr 2024 – Present

Current German Grade: tbd

- Courses: VLSI for DSP fundamental and architecture, FPGA, EDA, CAD, Advanced Compiler, Computer Arithmetic
- **VLSI Design Lab**: Designed and optimized a **D flip-flop** standard cell using Cadence Virtuoso with ASAP 7nm PDK, using Euler's Path Algorithm for compact and efficient layout. Performed **DRC**, **LVS**, and **PEX** checks with Calibre, and conducted timing and energy simulations using **Spectre** and Virtuoso Analysis Tools.
- Project: embedded processor design and optimization - used **ASIP-Designer GUI** to design sha256 algorithms on trv32p3 📄
- IDEA League Challenge Program 2024/2025: a leadership training program designed for outstanding students from 5 top universities (ETH Zurich, TU Delft, RWTH Aachen, Politecnico di Milano, Chalmers University).

University of Wisconsin-Madison, WI, USA

📄 Visiting International Student in Electrical and Computer Engineering

Sep 2023 – Dec 2023

Overall GPA: 4.0

- Courses: Digital System Design & Synthesis (📄Project: MazeRunner)
 - Designed and simulated key blocks like UART, PID, and SPI using Verilog with self-checking testbenches in ModelSim.
 - Developed synthesis scripts with **DC shell** for timing, PVT, and parasitic capacitance using Synopsys 32nm edu library.
 - Verified synthesized netlists via post-synthesis simulation and gained knowledge **IC Compiler** for APR implementation.
- Microprocessor Lab: PSoC4 MCU design, PCB PnR, and C programming for Whack-A-Mole game

National Cheng Kung University (NCKU), Tainan, Taiwan

B.Sc. in Electrical Engineering

Sep 2019 – Jan 2024

Overall GPA: 3.53 — Last 60% GPA: 4.0

- Award: Outstanding Student for Academic Achievements*1 (**Best GPA 4.3**)
- Courses: Electronics(A), Computer Organization(A), Data Structure, Digital Design, Digital IC Design 📄
- Clubs: president of NCKU Entrepreneur Association, member of NCKU Mountaineering Association

RESEARCH EXPERIENCE

Research Assistant (HiWi)

High Frequency Electronics, RWTH Aachen, Germany

Oct 2024 – Present

Advisor: Prof. Dr. Renato Negra

- Developed reusable FPGA libraries for UART TX/RX, SPI, DDR3 RAM interface, and Ethernet transfer.
- Built a modular system hierarchy, including a SPI arbitrator to select PLLs and DACs periphery on FD-DAC PCB.
- Designed a system for testing data storage in **DDR3 with read/write operations**, enabling autonomous high-speed data transmission to antennas via 8 channels on FD-DAC, and prepared for future taping out.
- Gained proficiency in **SystemVerilog** optimization, data communication protocols, and FD-DAC integration; collaborated effectively with German supervisors in English to achieve project milestones.

WORK HISTORY

Digital IC Designer Intern

FocalTech Systems

Jul 2023 – Aug 2023

Shenzhen, China

- Developed SRAM and register read/write functions for **XSI (eXternal Serial Interface)**, implementing FSM for command decoding and execution, and optimized burst and single data transfer modes.
- Created detailed command lists, state diagrams, and waveforms for **SPI operations** using Microsoft Visio.
- Resolved **cross-clock domain** challenges by designing synchronizers and dual-clock FIFOs for metastability mitigation, ensuring seamless data transfer between sclk and clk domains.

Software Engineer Intern

Wistron Corporation

Jan 2022 – Feb 2022

Kaohsiung, Taiwan

- Built a predictive module using **Python** and XGBoost, achieving 89.7% accuracy in turnover forecasts.
- Collaborated with SCRUM teams following **Agile methodologies**; Awarded **Pre-Employment Offer** as top 10 interns

SKILLS

- **Programming**: Verilog · C · TCL · Unix · Python · C++ · Matlab · git
- **Software**: Cadence Virtuoso · AMD Xilinx Vivado · SPICE · Synopsys EDA · ASIP-Designer · ModelSim · Altium Designer
- **Languages**: Chinese: Native English: Professional Proficiency German: Beginner
- **Hobbies**: Travel Photographer 📷, Internet Content Writer 📝